Introduction to Data Centre Management Using a Simulation Game

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Resumo The abstract should briefly summarize the contents of the paper in 150–250 words.

Keywords: Data centers simulation · Management in data centers · Simulation games · Game-Based Learning.

1 Introduction

The video game industry has experienced substantial growth over the past decade. In many countries, its economic impact has already surpassed that of traditional media sectors such as film, television, and newspapers. As the industry continues to expand in both scale and relevance, the application of video games increasingly extends beyond entertainment.

Serious games and simulation-based games provide players with interactive experiences that incorporate educational content. However, studies such as *Game-Based Learning with Computers: Learning, Simulations* [8] demonstrate that even games developed exclusively for entertainment require players to acquire new knowledge, develop complex skills, and devise effective strategies in order to progress through levels and overcome challenges.

Similarly, research such as *Video Games and Indirect Learning* by Monica Miller [7] suggests that the content of video games can also facilitate indirect learning, that is, learning that occurs without the game being explicitly designed for educational purposes.

Motivated by these findings, this work aims to investigate whether players acquire knowledge while interacting with **Port 0**, a resource management video game in which the player is responsible for operating a small data center within a fictional setting. Throughout the game, players must purchase, configure, and manage servers, fulfil client requests, and defend the infrastructure against cyberattacks.

Although the game is set in a science fiction universe, it incorporates several simulation-based mechanics, including word-based mini-games and server configuration tasks. The objective is to provide an engaging and enjoyable gameplay experience while simultaneously promoting knowledge acquisition, both through direct learning mechanisms embedded in the mini-games and through indirect learning arising from the contextual environment in which the player operates.

2 Related Work

O número de videojogos que transmitem conhecimento ao jogador, quer de forma direta ou indireta, é bastante alargado, podendo estes estarem englobados ou não na categoria de serius games. Em parelelo á industria de videojogos, diversos autores têm desenvolvido estudos e pesquisas sobre aprendizagem em videojogos, nos ultimos anos. *Trends* associadas á área de tecnologia como **data center** e **techonology** têm estado cada vez mais populadores nos ultimos anos.

Posto isto, ao longo desta seção o objetivo é analisa as estratégias de aprendizagem aplicadas ao videojogos e preceber como elas surtem efeito. Por fim analisar o crescimento das *trends* relacionadas com o contexto do jogo (**Port 0**).

2.1 Aprendizado por observação

Aprendizado por observação ou aprendizado por visualização é o processo pelo qual um iniciante absorve regras, passos iniciais e procedimentos complexos, simplesmente assistindo a jogadores experientes ou interagindo com simulações e/ou tutoriais [8]. Este tipo de aprendizagem quando integrado em sistemas de videojogos ou simulações atua como um "andeime"pedagógico para o aluno [8].

Alguns titulos como CodeCombat, Manufactoria, Labor Support Game, etc... Utilizam deste metodo para realizar o aprendizado.

No CodeCombat, por exemplo, os desenvolvedor utilizaram uma visão dividida onde o código é realçado linha a linha enquanto o herói executa as ações no ecrã, permitindo observar o resultado direto de cada comando [9].

Uma abordagem semelhante ao do CodeCombat foi tomada, no sentido de transmitir ao jogador o significado dos comandos, dentro do minigame (WordRush). Este minigame será abordado na subsection 3.2.

2.2 *Hype* pela área de tecnologia

3 Design and development of XX game

3.1 Design of XX game

3.2 Development of XX game

4 Results

5 Conclusion

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